Make a Summary

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1.

Four Methods for Communicating Enthalpies of Reaction

Method (typical units)	Symbol	Description
(1) Molar enthalpy of reaction (all in kJ/mol)	$\Delta_r H_m$ $\Delta_c H_m$ for complete combustion $\Delta_f H_m$ for formation from the elements $\Delta_{sd} H_m$ for simple decomposition	• The enthalpy change is stated for one mole of a substance undergoing a specified reaction. • A reaction equation is necessary only if the reaction is not one of the defined types. $\Delta_r H_m < 0$ (exothermic); $\Delta_r H_m > 0$ (endothermic)
(2) Enthalpy change (all in kJ)	Δ_rH $\Delta_c H$ for complete combustion $\Delta_f H$ for formation from the elements $\Delta_{sd} H$ for simple decomposition	• The enthalpy change is stated at the end of a given reaction equation. It must correspond to the balancing of the chemical equation. $\Delta_r H < 0$ (exothermic); $\Delta_r H > 0$ (endothermic)
(3) Energy term in a balanced equation (kJ)	(none)	• The quantity of energy involved is shown as a term in the balanced chemical equation. • on the reactant side for endothermic reactions on the product side for exothermic reactions
(4) Energy diagram (kJ)	E_p (chemical potential energy diagram)	• graph of E_p (no values) vs reaction coordinate $\Delta_r H$ labelled on graph with an arrow from reactants to products (up for endothermic; down for exothermic)
	$\Delta_f H°$ (enthalpy change diagram)	• graph of $\Delta_f H$ (with actual values on *y* axis) vs reaction coordinate • $\Delta_r H$ labelled on graph with an arrow from reactants to products (up for endothermic; down for exothermic)

Note that the symbol ° should be used only if the reactions are under standard conditions.

Three Methods for Determining enthalpy Changes

Method	Description	Key Assumptions
Calorimetry	$\underset{\text{calorimeter}}{Q} = mc\Delta t$ $\underset{\text{chemical}}{\Delta_r H_m} = \dfrac{\Delta_r H}{n}$	• No heat is transferred between the calorimeter and the outside environment. • A dilute aqueous solution has the same physical properties (density, specific heat capacity) as pure water. • Energy is conserved; i.e., the change in thermal energy is an indication of the change in enthalpy of the chemical system.
Hess' law	$\Delta_{net} H = \sum \Delta_r H$	• The reactions have the same initial and final conditions. The overall energy change for a net reaction is the same, regardless of the route followed to reach the products.
Standard molar enthalpies of formation	$\Delta_r H° = \sum n\Delta_{fP} H_m° - \sum n\Delta_{fR} H_m°$	• The standard molar enthalpies of formation of elements, in their standard states, are defined as zero.

2. 1. Photosynthesis is endothermic, cellular respiration is exothermic, and combustion of gasoline is exothermic.
   2. Coal-fired power stations burn coal, converting chemical energy to kinetic energy of steam. This kinetic energy is then converted to the kinetic energy of a turbine which, in turn, is converted to electrical energy in an electric generator. There are also natural gas power stations in Alberta, which involve similar energy transformations. Alberta also

produces some hydroelectricity. This involves energy conversions from potential energy of water (above the dam) to kinetic energy (falling water and spinning turbine) to electrical energy. Coal accounts for ~66% of the generated electricity in Alberta, gas for ~23%, and hydro for ~10%.

3. **Benefits of Coal**
   - plentiful
   - easy to convert to electrical energy (proven technology)
   - inexpensive

   Risks of Coal
   - causes pollution (produces NO_x and SO_2)
   - contributes to global warming
   - coal mining can result in environmental damage
4. solar energy → chemical energy (coal) → thermal energy (H_2O) → kinetic energy (steam moving the turbine that turns the generator) → electrical energy → light and heat
5. (a) If the reaction occurs in a particular medium (e.g., water), the change in temperature of that medium can be measured. This change in temperature is a reflection of the quantity of energy absorbed or released by the reaction in the medium, $Q = mc\Delta t$.
   (b) The energy released or absorbed by a chemical reaction is communicated by its enthalpy change, $\Delta_r H$, for a specified chemical equation, or by its molar enthalpy change, $\Delta_r H_m$, for a given substance reacting in a specified reaction.

Chapter 11 REVIEW

Part 1

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1. B
2. 1, 4, 3, 2
3. C
4. 585
5. D
6. 98.9
7. C
8. D
9. 1, 3, 6
10. 2, 4, 5
11. 3, 4, 2, 1
12. 3, 2, 1, 4
13. 1, 2, 4, 3
14. 136.4 *[Note error in text: should be labelled +]*

Solutions

4. $Q = mc\Delta t = 1.50 \text{ kg} \times \dfrac{4.19 \text{ J}}{\text{g} \bullet {}^\circ\text{C}} \times (98.6 - 5.5)\ {}^\circ\text{C} = 585 \text{ kJ}$

6. $\underset{SO_2}{\Delta_c H_m{}^\circ} = \dfrac{-197.8 \text{ kJ}}{2 \text{ mol}} = -98.9 \text{ kJ/mol}$

$\Delta_c H^\circ = \sum n\Delta_{fP} H_m{}^\circ - \sum n\Delta_{fR} H_m{}^\circ$ and $\Delta_c H^\circ_m = \dfrac{\Delta_r H^\circ}{n}$

12. $\Delta_c H^\circ_m$: $\underset{C_8H_{18}}{-5074.1 \text{ kJ/mol}}$; $\underset{C_3H_8}{-2043.9 \text{ kJ/mol}}$; $\underset{CH_4}{-802.5 \text{ kJ/mol}}$; $\underset{CH_3OH}{-637.9 \text{ kJ/mol}}$